

Han Bao

Shenzhen, China || Hangzhou, China (hometown)
Mobile: +86 18868702860
Email: hill_bao77@outlook.com
Personal Homepage: <https://broln7.github.io>
GitHub: github.com/BRoln7

Personal Profile

Robotics researcher focusing on embodied navigation and robot learning. My prior research aimed to tackle coupled motion planning and decision-making via reinforcement learning, while exploring implicit environmental perception through representation learning. Currently, I am developing navigation foundation models for safe, human-centric urban navigation. Additionally, my experience building a high-fidelity human-aware navigation benchmark has sparked a strong interest in RL-based character animation (DeepMimic, AMP) and humanoid control.

Education

2024 – present: MSc in Electrical and Electronic Engineering

Southern university of Science and Technology, Shenzhen

Supervised by Prof. Jiankun Wang.

Key Research Interests: Robot Learning, Embodied Navigation, Reinforcement Learning for Robotics.

2020 – 2024: B.Eng in Robotics

Zhejiang University of Technology, Hangzhou

GPA: 3.83/5.00 (Rank: 1/31).

Relevant Coursework: Introduction to Robotics: Mechanics and control, Modern Control Theory, Principle of Automatic Control, Signals and Systems, Machine Vision.

Thesis: Research on Robot Motion Planning Methods Based on Deep Reinforcement Learning in Dynamic and Complex Environments.

Publications

[1] Learning Robot Visual Navigation in Crowds via Intention-Aware Scene Representations

Han Bao, Bingyi Xia, Hanjing Ye, Yu Zhan, Hao Cheng, Baozhi Jia, Wenjun Xu, Jiankun Wang

IEEE Robotics and Automation Letters (RA-L 2026)

[Paper]

[2] NavIsaacLab: Generating Realistic Crowd via Parallel Robot Learning for Benchmarking Human-aware Navigation

Bingyi Xia*, Han Bao*, Jingyu Zhu, Hanjing Ye, Yuhan Pang, Guangcheng Chen, Liang Lin, Zhengyou Zhang, Wenjun Xu, Jiankun Wang

IEEE Transactions on Automation Science and Engineering (T-ASE 2026) (*equal contribution)

[Preprint]

[3] Exposing the Long-tail in Embodied Urban Navigation via Scalable Learning from In-the-Wild Videos

Bingyi Xia, Han Bao, Zhewei Chen, Hanjing Ye, Jingwen Yu, Yuhan Pang, Wenjun Xu, Jiankun Wang

Under review

[Preprint]

Projects/Research

Robot Urban Navigation Foundation Model | 2025.12-now

Shenzhen Key Laboratory of Robotics Perception and Intelligence (r π Lab), SUSTech

- We extend prior research on human-aware robot navigation to more general scenarios, aiming to develop efficient and generalizable Physical AI that aligns with humans in complex urban environments. [Homepage]

Human-Aware Robot Navigation | 2024.03-2025.09

Shenzhen Key Laboratory of Robotics Perception and Intelligence (r π Lab), SUSTech

- Developed a high-fidelity human-aware robot navigation benchmark platform based on IsaacLab. It generates crowd behaviors with realistic motions using character animation control methods, and provides multi-scale real-world scene assets along with baseline navigation algorithms. [Homepage]
- Designed a novel robot visual navigation method in crowds that utilizes BEV (Bird's Eye View) encoding and human pose embedding technologies to derive intention-aware scene representations. This representation significantly enhances the training efficiency of DRL-based robot navigation, achieving state-of-

the-art (SOTA) performance. [\[Homepage\]](#)

Vision-Guided Pan-Tilt Laser System | 2023

First Prize of Zhejiang Province, National Undergraduate Electronics Design Contest

- Developed a machine-vision-based pan-tilt control system that actuates a laser pointer for precise tracing of black tape boundaries and real-time dynamic tracking of a moving colored laser spot.

Vision-Guided Self-Balancing Bicycle | 2022

National First Prize, 17th National Undergraduate Smart Car Contest (Balancing Bicycle Category)

- Engineered an autonomous self-balancing bicycle system driven by a reaction wheel for active posture control. By integrating a real-time machine vision pipeline for precise trajectory tracking, the platform achieved robust dynamic balancing and high-speed racing. [\[Demo\]](#)

Honors and Awards

- | | |
|--|------------|
| • Outstanding Undergraduate Graduate of Zhejiang Province | 2024 |
| • Zhejiang Provincial Government Scholarship | 2022, 2023 |
| • First Prize (Zhejiang Province), National Undergraduate Electronics Design Contest | 2023 |
| • National First Prize, 17th National Undergraduate Smart Car Contest | 2022 |

Skills

- Programming: Python, PyTorch, C/C++
- Robotic: ROS/ROS2, NVIDIA Isaac-Sim/IsaacLab

Academic Service

Journal & Conference Reviewer

IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Automation Science and Engineering (T-ASE)
IEEE International Conference on Robotics and Automation (ICRA)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)